

Pocket Vakil

A domain-specific SLM for Indian criminal law — predicting applicable IPC sections directly from case facts.

332M PARAMETERS | 13,000+ CASES | BUILT FROM SCRATCH

DOMAIN-SPECIFIC SLM EVALUATION — TEAM INTAKE

Model, Architecture & Domain Details

MODEL & ARCHITECTURE

Model name / version
Pocket Vakil v2 (HF: ORG/pocket-vakil-332m-ipc)

Base model or from-scratch
From scratch (OLM 2.2.0 primitives, no pretrained weights)

Exact parameter count (≤1B)
331,964,416

Attention type
MHA / GQA / MQA
MHA — full-width q/k/v projections, 16 heads, head_dim 64, no grouping

Normalization
RMSNorm / LayerNorm
LayerNorm

Activation
SwiGLU / GeLU
SwiGLU, ff_multiplier 2.625

Positional encoding
RoPE / ALiBi / learned
RoPE

Context length
2048

Precision
fp16 / bf16 / int8
bf16

Tokenizer
Vocab size · custom vs existing · domain-specific additions
Custom BPE, 32,000 vocab, trained on the domain corpus. Domain additions present at tokenizer-training time: 100 IPC label tokens

Chat template / prompt format
<passage>\n\n### TASK: <instruction>\n### ANSWER:\n

DOMAIN & TASK

Domain name
Indian criminal law (Indian Penal Code)

Primary task type
QA / classification / generation / other
Classification — multi-label

End-user & use-case description
Legal researchers, practitioners and paralegals. Given case facts, the model surfaces applicable IPC sections as a candidate set for human review. Designed to assist statute identification, not to replace legal judgement.

Data, Training, Compute & Reproducibility

DATA

Training data source(s) & size
Tokens / examples

Pretraining data: Open India Law, ~2.0B tokens, 976,379 sequences across 49 shards.
Coverage: Supreme Court + 7 High Courts, with Bombay/Madras capped to avoid court-level dominance.
Fine-tuning: ILSI (Paul et al., AAAI 2022), 52,750 examples.
IPC: 42,750 (81.0%)
Instruction: 6,000 (11.4%)
Continuation: 4,000 (7.6%)

Data license / collection method

Public court judgments, publicly available. Collected and processed via src/build_corpus.py. ILSI used under its published terms.

Train / val / test split sizes

ILSI as distributed — train 42,750 / dev 10,181 / test 13,019. 100-label set. Test held out of all training and tuning.

Contamination check method

Test-set deduplication approach

Test set decontamination approach

Training corpus (or file-level hashes)
Required for Level 3 n-gram contamination check against hidden benchmark

Per-shard SHA256 not computed before the compute node was reclaimed at window close. Corpus is reproducible from public sources via src/build_corpus.py with data/shards/manifest.json, which records exact per-court token counts, caps applied, documents dropped, and the tokenizer hash

TRAINING & COMPUTE

Optimizer, LR & schedule
Warmup steps, decay type

Optimizer: AdamW, β=(0.9, 0.999), ε=1e-8, weight decay 0.1, gradient clipping 1.0.
LR Schedule: Warmup + cosine decay.
Slot 1: 6e-4 peak LR → 6e-5, 500 warmup, 4,964 steps.
Slot 2: 2e-4 peak LR → 6e-5, 100 warmup, 5,086 steps,

Effective batch size · epochs/steps

micro-batch 12 × grad-accum 16 = 192 sequences/step = 393,024 tokens/step (192 × 2047; targets are seq[1:]).

GPU type & total GPU hours

1 × NVIDIA H100 NVL (93 GB), 18.6 hours

tokens_seen
Must match manifest.yaml & train_log

3,945,960,960 at the final logged step (10,040); 3,930,240,000 at step_10000.pt, the checkpoint the submitted model was fine-tuned from. 11.8 tokens/parameter — ~60% of Chinchilla-optimal.

Random seed(s) used

1234

Post-training: SFT performed?
Method (full fine-tune / LoRA), dataset

SFT: Yes
Dataset:52,750 examples
42,750 IPC (81.0%)
6,000 instruction-following (11.4%)
4,000 continuation (7.6%)
Method: Full fine-tuning, no LoRA
Initialization: step_10000.pt

What Pocket Vakil Proves

1

Domain Specialization Wins

A compact, purpose-built SLM can outperform larger general systems on a focused legal task.

2

Efficiency Is Achievable

332M parameters, 18.6 GPU-hours, one GPU — competitive results don't require massive compute.

3

Data Quality Over Quantity

The v2 fine-tuning lesson proves that the right mixture of task-framed data beats simply adding more instructions.

4

Local, Transparent Evaluation

Supports legal research without dependence on large external APIs.

Results: Competitive with Published Systems

Beats nearly every published baseline on Macro-F1 — second only to LADAN-L. Achieved with a 332M from-scratch model on a single GPU, not a large fine-tuned system.



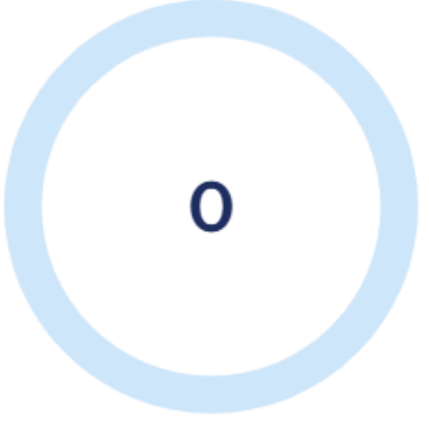
Micro-F1

(within 0.007 of AoS, the strongest published baseline)



Macro-F1

(outperforms TF-IDF, Logistic Regression, LeSICiN, and AoS)



Dead Labels

out of 100 (all IPC sections achieved non-zero F1)



Parse Failures

(near-perfect output reliability)

Conclusion

Main Contributions

- Built large-scale Indian legal corpus pipeline with SOTA results
- Trained domain-specific 32K tokenizer
- 332M Transformer trained from scratch
- Mixed-task fine-tuning strategy
- Multi-label IPC evaluation framework

Limitations & Responsible Use

- Not a legal advice system
- Predictions require professional verification
- Benchmark scores $\neq$ legal correctness

Future Work

- Dedicated multi-label classification head
- Increase pretraining token budget
- Longer context windows
- Label threshold tuning
- Improve rare-label performance
- Add model explanations